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Ethanol condensation to butanol at high temperatures over a basic heterogeneous catalyst: How relevant is acetaldehyde self-aldolization?

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ABSTRACT

The condensation of ethanol to butanol was investigated over a commercial hydroxyapatite catalyst in the 350–410 °C temperature range. An analysis of thermodynamic and kinetic data, including the measure of the concentration of water and dihydrogen formed during the reaction, unambiguously revealed that the pathway involving acetaldehyde self-aldol condensation is irrelevant at such high temperatures for the present catalyst. At least two reaction pathways are suggested to take place simultaneously. The main pathway would involve the condensation of two ethanol molecules with apparently no intermediate gaseous compounds (so-called "direct" route). A minor "indirect" route would involve the condensation of ethanol with acetaldehyde (formed from ethanol dehydrogenation) to form butenol, which is subsequently converted to butanol by hydrogen transfer from a sacrificial ethanol molecule. This minor route would be less selective, resulting in the formation of acetaldehyde and H_2 as by-products. The alcohol condensation mechanism(s) taking place over basic oxides at high temperatures would therefore be fundamentally different from that taking place over bi-functional solids (containing both metallic and basic sites) at lower temperatures. In a more general context, this work underlines the benefits of considering thermodynamic data when assessing the relevance of potential reaction pathways.

1. Introduction

Ethanol, a renewable chemical, can be converted to higher molecular weight alcohols, so-called Guerbet alcohols [1,2], to be used, for instance, in the production of high-added value solvents and surfactants. The traditional (partly-homogeneous) synthesis of Guerbet alcohols is known to proceed via several consecutive steps [3,4]. Aldehyde formation by alcohol dehydrogenation and its self-aldolization is a well-accepted reaction pathway in the case of the reaction catalyzed by alkali metal hydroxides in the presence of a metallic catalyst.

In the case of the reaction starting from ethanol, the metal carries out the ethanol dehydrogenation to acetaldehyde, as well as the hydrogenation of reaction intermediates to butanol (Scheme 1). The homogeneous base is responsible for acetaldehyde self-aldolization to 3-hydroxybutyraldehyde, which is then dehydrated to crotonaldehyde. Crotonaldehyde hydrogenation eventually leads to butanol. It is clear that a purely heterogeneous process would present many advantages over the hybrid homogeneous-heterogeneous system aforementioned.

The conversion of ethanol to butanol over purely heterogeneous catalysts has been reported in several publications and patents [2,5–15]. Basic oxide-supported metals and especially non-promoted (i.e. metal-free) basic oxides were shown to be active and selective, with hydroxyapatites presenting some of the highest selectivity to higher alcohols [6,7,13].

Several studies have addressed the nature of the reaction mechanism over purely heterogeneous basic solids. The relevance of the self-aldolization of acetaldehyde (second step in Scheme 1) has been widely discussed. A number of authors have proposed that acetaldehyde self-aldolization was a crucial reaction step [6–9]. In contrast, a direct reaction between two ethanol molecules was proposed on various basic solids such as a multicomponent oxide catalyst MgO—CuO—MnO [10], alkali-exchanged zeolites [11], MgO [12] and hydroxyapatites [13]. Iglesia and co-workers proposed that the conversion of ethanol to butanol could be achieved by two main reaction pathways, one direct and one via acetaldehyde self-aldolization [14]. These conclusions came from the facts that butanol was found to be both a primary and secondary reaction product.

In the present contribution, the concentration of the main reaction products, including for the first time water and dihydrogen, was monitored over a commercial hydroxyapatite (noted HAP). Surprisingly, our results will rigorously show that a mechanism

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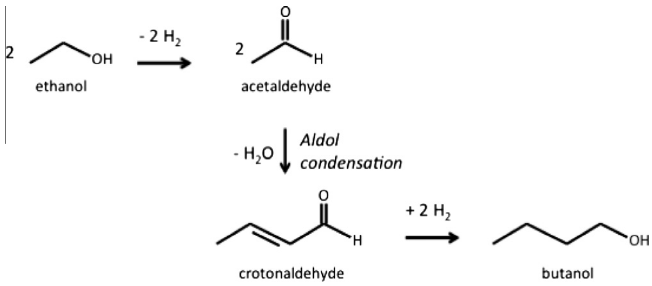


Scheme 1. Ethanol condensation mechanism based on the self-aldolization of acetaldehyde.

based on acetaldehyde self-aldolization (as described in Scheme 1) has a negligible contribution to butanol formation, and that at least two other reaction pathways occurring simultaneously must be considered in our case.

2. Experimental

Ethanol (from Prolabo 99.5%, the main impurity being water) was fed using a saturator kept at 45 °C into a heat-traced stainless steel flow setup. The catalyst was placed in a quartz tube reactor and held between quartz wool plugs. The reactor was located in a tubular furnace. Ar was used as carrier gas. A combination of mass spectrometry, gas-chromatography and gas-phase FT-IR spectroscopy was used to determine the exact concentration of ethanol and the main products of interest (including H_2O and H_2) at the reactor exit.

The gas chromatograph (Bruker 450-GC) was fitted with Zebron ZB-BIOETHANOL column (30 m, 0.25 mm, film thickness: 1.00 μm). A flam ionization detector (FID) was used and a precise quantification of all the detectable products was realized through the use of an internal standard (i.e. toluene) added to the analyzed stream before injection in the column. The concentration of most reaction products, including dihydrogen, could also be monitored by on-line mass spectrometry (Pfeiffer Omnistar 320). The contribution of large molecules to the fragment $m/z = 2$ was taken into account to determine the H_2 concentration, also using calibration curves. The concentration of some reaction products, in particular acetaldehyde and water, was also monitored by on-line FT-IR gas analysis using a 27 cm-long single path gas cell fitted in a Nicolet 560 spectrometer. Calibration curves were drawn to relate IR band signal intensity to concentrations.

The hydroxyapatite (82 $m^2 g^{-1}$), supplied by Acros Organics (Batch A0312711), was activated under Ar at 480 °C for 1 h before introducing the ethanol/Ar feed at the reaction temperature. The thermodynamic calculations were done with the HSC Chemistry®software (version 6.2, by Outotec).

3. Results and discussion

3.1. Kinetic data

The activity of the HAP was measured over the 350–440 °C temperature range (Fig. 1). Butanol was always the main reaction product and acetaldehyde the main by-product. The conversion was always lower than 30% and the selectivity to C_{6+} products was negligible. The present HAP presented a lower selectivity to butanol (ca. 50%) than that of the HAP samples reported by Ueda and co-workers [7], possibly due to differences in the reaction conditions and/or sample composition and texture. It must be stressed that the point of the present contribution was not to improve on the highest reported selectivity to butanol, but to study a reference sample that displayed acceptable activity/selectivity patterns with respect to those measured on the many basic oxides that have been used for ethanol condensation [6–14].

The evolution of the concentration of butanol and acetaldehyde was followed as a function of the contact time (more precisely, the catalyst weight to ethanol molar flow ratio) at 400 °C (Fig. 2). The yield of acetaldehyde was initially proportional to the contact time before leveling off. This suggests that acetaldehyde was a primary reaction product, which led to secondary reaction products at higher conversions.

The yield of butanol was represented by a curve with a complex shape. The initial slope was different from zero, suggesting that the butanol was formed as a primary reaction product at low contact times. The production of butanol then increased more rapidly before showing again a monotonous increase. This indicates that some of the butanol was then also formed as a secondary reaction product. The sudden increase in butanol corresponded to the point at which acetaldehyde was leveling off, suggesting that part of the acetaldehyde was converted to butanol. The slope and offset of the lines representing butanol concentration in the limiting cases of low and high contact times (Fig. 2) suggest that most of the butanol was always formed via the direct “primary” reaction pathway.

Similar butanol concentration versus contact time curves had been observed by Iglesia and co-workers in the case of ethanol conversion over Mg—Al mixed oxides [14], which had also led these authors to propose that butanol was formed through more than one reaction pathway.

The facts that (i) the slope of the two lines relating to the formation of butenol at low and high W/F appears to be identical and (ii) the butanol formation shows an offset above a certain W/F are worth noting. A possible explanation of these observations is that

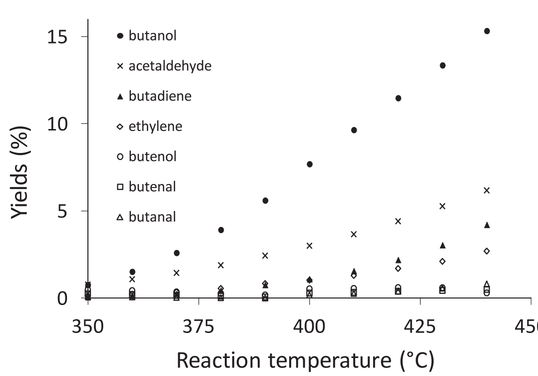


Fig. 1. Yields of the main C-containing reaction products during ethanol reaction over the hydroxyapatite as a function of temperature. Butenol refers to 2-buten-1-ol and 3-buten-1-ol. Feed: Ethanol = 15.2% in; WHSV = 14 h^{-1}.

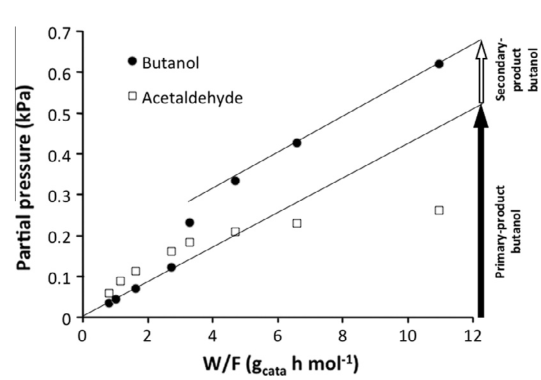


Fig. 2. Butanol and acetaldehyde partial pressure at the reactor exit at 400 °C for various W/F values. Ethanol concentration = 7.6%.

the reaction site at which butanol is formed from acetaldehyde tends to deactivate at higher contact time. This is possible since acetaldehyde is highly reactive and can lead to cyclic compounds (via self-aldolization) favoring coke formation. This explanation is supported by the fact that acetaldehyde production tends to level off over our HAP (Fig. 2) and a similar acetaldehyde leveling off can be noted in the case of the data reported by Di Cosimo et al. [14] over a Mg—Al—O catalyst. This hypothetical deactivation of the sites involving acetaldehyde would suggest that the corresponding reaction pathway would remain minor and its overall contribution to the formation of butanol could even decrease with time on stream.

3.2. Thermodynamic analysis

Butanol and acetaldehyde are the main carbon-containing reaction products over many ethanol condensation catalysts as the one investigated here. It is yet quite surprising that no study has ever reported any thermodynamic considerations on the relative concentrations of these two compounds, to the best of our knowledge. Such analyses can be useful in discarding some potential reaction steps, for instance all those associated with a reaction quotient (noted "Q") higher than the corresponding equilibrium constant (noted "K"). This is equivalent to having an approach to equilibrium η (=Q/K) higher than 1. This approach has proved successful in unraveling the mechanism of methanol formation over Cu-based catalysts [16] and NO oxidation to NO_2 over alumina-based catalysts [17].

The comparison of the reaction quotient and equilibrium constant requires the determination of the concentration of all the species involved in the reaction step. Crucially, water and dihydrogen concentrations were measured here, since these compounds take part in the reaction step of interest. An example of the calculation of a reaction quotient is detailed in the supporting information.

The simplest associated global reaction that can be computed for acetaldehyde self-aldolization (with intermediates of accurately measurable concentrations and available thermochemical data) is given by reaction (R1):

$$2\ \text{acetaldehyde} + 2H_2 \leftrightarrows \text{butanol} + \text{water} \qquad (R1)$$

The corresponding thermodynamic equilibrium graph is shown in Fig. 3 and the equilibrium constant at 400 °C is given in Table 1. Importantly, Fig. 3 shows that the reactant side (acetaldehyde and H_2) is favored at the higher temperatures, due to its higher entropy (as more gaseous molecules are present on the reactant side). In fact, if only reaction (R1) were taking place, the concentration of butanol would be expected to remain below that of acetaldehyde

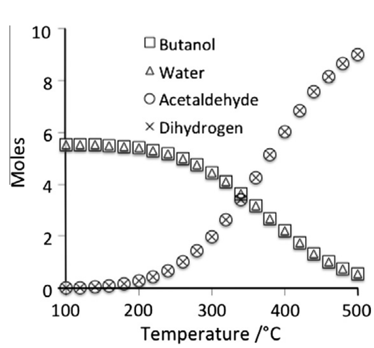


Fig. 3. Composition at the thermodynamic equilibrium of a system containing acetaldehyde, H_2, butanol and water as a function of temperature. Total pressure is 1 bar and the initial state of the system corresponds to an equivalent 11 mol of ethanol (balanced by Ar).

above 350 °C. This is clearly not the case for the data collected at 400 °C (Fig. 1) and a W/F higher than 3 g h mol^{-1} (Fig. 2), for which the butanol concentration was larger than that of acetaldehyde.

Since other parallel or consecutive reactions might produce or consume the compounds involved in R1, the reaction quotient Q must be exactly calculated and compared to the corresponding equilibrium constant K. The reaction quotient Q determined at 400 °C was found to be ca. 8.8×10^4, that is, a value several orders of magnitude higher than the value (i.e. 36.4) of the equilibrium constant K associated with R1. Therefore, it can be concluded that the contribution of R1 was negligible as an intermediate reaction step (involving gaseous acetaldehyde and H_2) in the global pathway converting ethanol to butanol over our HAP. In other words, the reaction pattern as shown in Scheme 1 was highly insignificant and one or more other routes to butanol did prevail.

Values of the reaction quotient were measured under many other conditions and temperatures between 350 and 410 °C and were always found to be orders of magnitude higher than the corresponding equilibrium constant, even when the experimental errors associated with concentration measurements were cumulated at their maximal value (Table S1, Supporting Information). Therefore in all these cases, the overwhelming fraction of butanol was not formed from ethanol via a reaction scheme involving the reaction (R1) that implies acetaldehyde self-aldolization.

This conclusion is important, because it unambiguously shows that the reaction pathway occurring over our HAP catalyst under our experimental conditions does not proceed like the metal-promoted partly homogeneous reaction scheme involving acetaldehyde self-aldolization. Note that this partly homogeneous process is typically run at 220 °C and therefore butanol formation is essentially not thermodynamically limited (Fig. 3).

Thus, it is reasonable to propose that the low-temperature partly homogeneous metal-promoted alcohol condensation process and the high-temperature fully heterogeneous reaction occurring on non-promoted oxides could be fundamentally different in nature. As a matter of fact, oxides such as HAP or MgO should be less efficient in catalyzing simultaneously both (i) ethanol dehydrogenation and (ii) C_4-intermediate hydrogenation than Raney Ni or Cu metals, typically used as promoters in the case of the partly homogeneous reaction [3,4]. The hydrogenation step involves surface H atoms in the case of the metals, while hydrogenation of the C_4-intermediates is more likely to occur via hydrogen transfer between two molecular species in the case of plain oxides.

We would like to emphasize again that our present combination of detailed reaction product analysis and thermodynamic considerations unambiguously proves that, precisely, acetaldehyde self-aldolization is not part of a main reaction pathway over our HAP under the range of experimental conditions used. Note that it does not mean that acetaldehyde is not part of a reaction mechanism, i.e. this molecule could possibly react with ethanol, for instance, to lead to a C_4-intermediate, and this will be discussed in the subsequent section. The data reported in the present section are therefore a major advancement from the inspiring earlier studies that reported inconsistencies with the role of acetaldehyde as reaction intermediate. It would be interesting to carry out similar

Table 1
Theoretical equilibrium constant and measured reaction quotient for the conversion of acetaldehyde and dihydrogen into butanol and water at 400 °C. Reaction conditions: ethanol = 15.2%, W/F = 3.35 g h mol^{-1}; WHSV = 28 h^{-1}.

Reaction	Equilibrium constant K	Reaction quotient Q
2 Acetaldehyde + 2 H_2 = butanol + water	36.4	8.8×10^4

measurements over other catalytic formulations used at similar temperatures. The thermodynamic considerations developed here may yet not be useful for systems operating at low temperatures, since the associated reaction constants imply high butanol/acetaldehyde ratios (Fig. 3).

Another advantage of the thermodynamic analysis is that the data it is based upon were collected under reaction condition at constant ethanol pressure, therefore not involving transient regimes that may lead to different surface states and reactions as compared to the case of true steady-state operation.

3.3. Potential reaction pathways consistent with thermodynamics

The kinetic data reported in Fig. 2 suggest that a major fraction of the butanol was formed as a primary reaction product, according to a "direct" route:

$$2\ \text{ethanol} \rightarrow \text{butanol} + \text{water} \qquad \text{(R2)}$$

One cannot totally rule out that there may be intermediate compounds (i.e. species that effectively desorbed from the catalyst surface) and butanol formed by R2 may not be a truly primary product, but those elusive intermediates would then be very short-lived and possibly not measurable by standard methods. The thermodynamic equilibrium plot associated with ethanol, water and butanol indicates that butanol and water are by far the most stable products (Fig. S1) and, clearly, such a direct scheme cannot be ruled out based on thermodynamic grounds. We unfortunately do not have any suggestions as to the details of such direct pathway, which had been already proposed in previous Refs. [10–14] and mentioned in a recent review [18].

The kinetic data reported in Fig. 2 also suggest that a minor fraction of butanol could be formed via acetaldehyde as an intermediate. Since it was rigorously proven in the former section that acetaldehyde self-aldolization was irrelevant, the simpler alternative coupling that could be proposed is the reaction between one molecule of acetaldehyde and one molecule of ethanol:

$$\text{Ethanol} + \text{acetaldehyde} + H_2 \rightarrow \text{butanol} + \text{water} \qquad \text{(R3)}$$

Such reaction scheme had been discussed by Yang and Meng [11], who stressed that this reaction pathway (R3) over their Rb-NaX catalyst was minor in comparison with the direct route (R2). These authors proposed that ethanol and acetaldehyde would condense to butanal, which would subsequently be hydrogenated to butanol. However, the intermediacy of butanal is questionable, if one considers the thermodynamic equilibrium of a system involving these compounds. Without going into detailed calculations of reaction quotients, it is obvious that butanal should remain the major C-containing product, if it were formed as an intermediate (Fig. S2). Yet, the measured butanal concentration remained significantly lower than those butanol and other products (Fig. 1). In fact, butanal would more likely be formed from the dehydrogenation of butanol, as proposed by Yang and Meng [11] and Ndou et al. [12], who showed that butanol can be converted to butanal over basic oxides.

Other compounds that could be formed by the reaction of acetaldehyde and ethanol are butenols (e.g. 3-buten-1-ol and 2-buten-1-ol), which can be subsequently hydrogenated to butanol:

$$\text{Ethanol} + \text{acetaldehyde} \rightarrow \text{butenol} + \text{water} \qquad \text{(R4)}$$

$$\text{Butenol} + H_2 \rightarrow \text{butanol} \qquad \text{(R5)}$$

The yields of butenols were actually small in the present study and the error associated with their concentration did not allow calculating a reaction quotient with a satisfactory precision. Nevertheless, the thermodynamic equilibrium data pertaining to a system comprising butanol, 3-buten-1-ol, 2-buten-1-ol and H_2 indicate that achieving butanol concentrations markedly higher than those of butenols is possible (Fig. 4) and therefore the reaction pathway R4 + R5 cannot be rejected. The formation of butadiene, which is a significant by-product (Fig. 1), could be easily explained by the dehydration of 3-buten-1-ol. The reaction (R4) could first involve the condensation of ethanol with the tautomeric form of acetaldehyde, ethenol, leading to a mixture of butenols (Scheme 2, left).

This route based on acetaldehyde and ethanol could possibly involve the formation of an intermediate diol (R6), which would lead to the butenol by dehydration (R7). The expected equilibrium concentrations of butanediol are significantly smaller than those of butenols (Fig. S3) and therefore butanediol could be hardly detectable experimentally.

$$\text{Ethanol} + \text{acetaldehyde} \rightarrow \text{butanediol} \qquad \text{(R6)}$$

$$\text{Butanediol} \rightarrow \text{butenol} + \text{water} \qquad \text{(R7)}$$

To support further the potential role of the reaction between acetaldehyde and ethanol, the reaction quotient associated with the reaction (R8) was calculated (see Supplementary Information, Table S2). The reaction quotient was always significantly lower than the corresponding equilibrium value, indicating that reaction (R8) cannot be ruled out.

$$\text{Ethanol} + \text{acetaldehyde} + H_2 \rightarrow \text{butanol} + \text{water} \qquad \text{(R8)}$$

A comment should be made on the final reaction step consisting in the hydrogenation of butenols (R5). Similarly to other systems based solely on basic oxides, hydroxyapatite does not contain a metallic phase that would readily hydrogenate the C=C double bond of butenols. Apesteguia and co-workers [19] have already proposed that the final hydrogenation could be realized via a hydride transfer according to a Meerwein–Ponndorf–Verley mechanism. The hydrogen donor would be a (sacrificial) ethanol molecule, leading to acetaldehyde as by-product:

$$\text{Butenol} + \text{ethanol} \rightarrow \text{butanol} + \text{acetaldehyde} \qquad \text{(R9)}$$

In summary, alongside the main "direct" condensation reaction (R2), we would like to propose the following reaction mechanism as the minor "indirect" route to butanol (R11), based on the data reported in the literature over other basic oxides and the kinetic and thermodynamic considerations reported in the present work:

$$\textit{Step}\ 1: \text{Ethanol} \rightarrow \text{acetaldehyde} + H_2 \qquad \text{(R10)}$$

$$\textit{Step}\ 2: 2a: \text{Ethanol} + \text{acetaldehyde} \rightarrow \text{butenol} + \text{water} \qquad \text{(R4)}$$

or

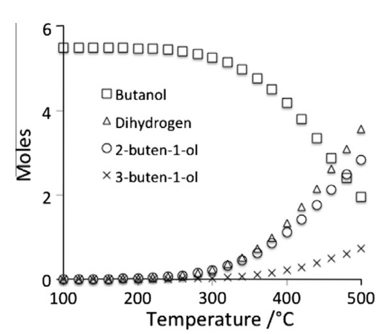
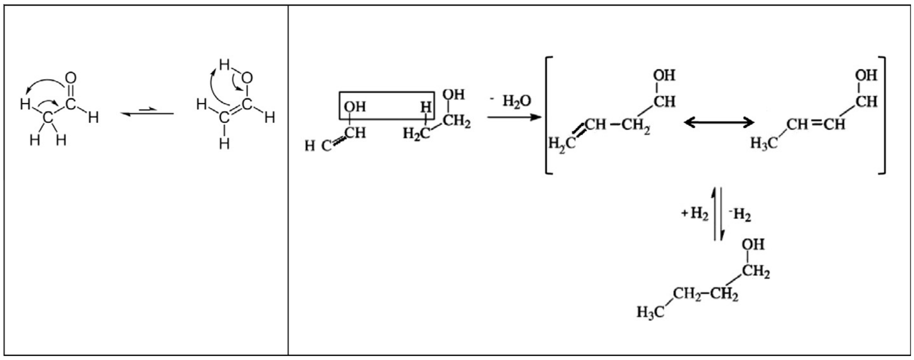


Fig. 4. Composition at the thermodynamic equilibrium of a system containing butanol, 2-buten-1-ol, 3-buten-1-ol and H_2 as a function of temperature. Total pressure is 1 bar and the initial state of the system corresponds to an equivalent 11 mol of ethanol (balanced by Ar).



Scheme 2. Butanol formation based on the condensation of acetaldehyde and ethanol, via butenols. Left: the first step would be the tautomerization of acetaldehyde to ethenol. Right: butenols are formed as reaction intermediates.

$$2b1: \text{ Ethanol} + \text{acetaldehyde} \rightarrow \text{butanediol} \tag{R6}$$

$$2b2: \text{ Butanediol} \rightarrow \text{butenol} + \text{water} \tag{R7}$$

$$\textit{Step } 3: \text{ butenol} + \text{ethanol} \rightarrow \text{butanol} + \text{acetaldehyde} \tag{R9}$$

$$\text{Overall}: \text{ 3 ethanol} \rightarrow \text{butanol} + \text{water} + H_2 + \text{acetaldehyde} \tag{R11}$$

The reaction quotient was calculated for the reaction (R11) and was found to be several orders of magnitude lower than the corresponding equilibrium constant, showing that this global reaction cannot be rejected on thermodynamic grounds (see Supplementary Information, Table S3).

It is obvious that more work is needed to ascertain the relevance of the "minor" reaction scheme R11 and especially to determine the details of the main mechanism R2. It should be noted that R11 has an intrinsic selectivity limit (i.e. 2/3 of the ethanol molecules end up as butanol), with acetaldehyde and H_2 being side-products. Since the acetaldehyde produced by R9 can be recycled in R4 or R6 (resulting in a propagation-like mechanism), the effective selectivity associated with R11 will be higher than 2/3 and will also depend on the experimental conditions affecting acetaldehyde readsorption. The theoretical selectivity of the direct route R2 is 100% and it would therefore be crucial to better understand the active sites associated with it, in particular to determine whether this site only catalyzes R2 or is also responsible for other reactions.

Since H_2 chemical readsorption is hardly expected on this type of metal-free solid, it is likely that any H_2 formed will leave the reactor as such. It should also be noted that the acetaldehyde molecules formed by ethanol dehydrogenation resulting in H_2 production (R10) and those formed via hydrogen transfer to an unsaturated compound (R9) possibly occurred on different sites. It would be interesting, for instance, to study HAP with different morphologies and to determine whether or not the initial selectivity to butanol can be increased.

4. Conclusions

The reaction pathway involving acetaldehyde self-aldolization is irrelevant for the condensation of ethanol to butanol in the 350–410 °C temperature range over the present commercial hydroxyapatite catalyst. The presence of at least two other reaction routes to butanol is suggested. The main suggested "direct" route probably involves the condensation of two ethanol molecules. A minor "indirect" route probably involves the condensation of ethanol with acetaldehyde (produced from ethanol dehydrogenation) to form a butenol, which is eventually converted to butanol by hydrogen transfer from a sacrificial ethanol molecule. The minor route is expected to be less selective and result in the formation of H_2 and acetaldehyde as by-products.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.jcat.2013.11.004.

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